Morphological processing in the parafovea for deaf and hearing readers

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The earliest stage of word recognition during sentence reading occurs while an upcoming word is in the reader's parafovea, between 2 and 5º of visual angle from the central fixation. Successfully preprocessing information from the upcoming word, such as morphological structure, can facilitate readers' processing of the target word once they fixate on it. These patterns have been observed in languages such as German and Hebrew (e.g., Mousikou & Schroeder, 2019; Deutsch et al., 2000). Evidence of parafoveal preprocessing of morphology during sentence reading in English is mixed, however; readers demonstrate a preview benefit from morphological structure for suffixed words, but not compound or prefixed words (Dann et al., 2021; Kambe, 2004). These results suggest that readers of English only preprocess morphological structure when the more salient stem (as opposed to the less semantically informative prefix) falls in the highest acuity area of the parafoveal region directly next to the central fixation (Dann et al., 2021).

One group of English readers who may differ in this pattern is those who are prelingually deaf. These readers are able to attend to orthographic information further into the periphery during sentence reading than hearing readers (Bélanger et al., 2012), which may contribute to their efficient patterns of eye movements (see Traxler et al., 2021). Morphological awareness (MA) also has a stronger role in reading comprehension for deaf readers (Saunders et al., 2025), but little is known about how it affects online sentence processing. The combination of these patterns may result in an enhanced ability to pre-process morphological structure for deaf readers compared to their hearing peers.

Using a gaze-contingent display change paradigm, we tested whether deaf and hearing readers with varying MA showed differences in parafoveal processing of morphology. Participants (N=48) read English sentences as their eye movements were recorded. Target words consisted of a stem and a suffix (*sadness*), while preview words either had a pseudomorphological suffix (*sadment*), a nonmorphological suffix (*sadnard*), or were unrelated nonwords (*florous*). Following Dann et al., (2021), both deaf and hearing readers were hypothesized to show a graded preview benefit, with the most benefit from pseudomorphological previews, then nonmorphological, then unrelated nonword previews. Deaf readers were also hypothesized to show a larger preview benefit due to enhanced sensitivity to morpho-orthographic structure. Linear mixed effects models revealed a three-way interaction between deafness, MA, and preview condition (p = 0.04). Deaf readers with higher MA showed a graded priming effect, with shorter gaze durations on target words following a pseudomorphological preview compared to a nonmorphological preview. Hearing readers were unaffected by the morphological preview. These results suggest that deaf readers are more attuned to morphological structure in the parafovea during sentence reading, but only if they have a higher level of MA, aligning with previous findings that suggest that MA has a stronger relationship with reading for deaf readers at higher skill levels (Saunders et al., 2025).

The current study found that during sentence reading, deaf readers with strong morphological awareness carry out initial morpho-orthographic segmentation processes in the parafovea faster than both hearing readers and deaf readers with weak morphological

awareness. This enhanced ability to attend to morphological structure in the parafovea facilitates recognition of complex multimorphemic words, allowing these deaf readers to more efficiently allocate their attention (i.e., their fixations) to relatively long, infrequent words that may require more processing resources.

Preview Condition	Example sentence
Identity:	The man tried hiding his | sadness from his best friends.
PseudoMorphological nonword:	The man tried hiding his | sadment from his best friends.
NonMorphological nonword:	The man tried hiding his | sadnard from his best friends.
Unrelated nonword control:	The man tried hiding his | florous from his best friends.

Figure 1. Example sentences with each type of preview condition. Vertical line indicates the location of the invisible boundary.

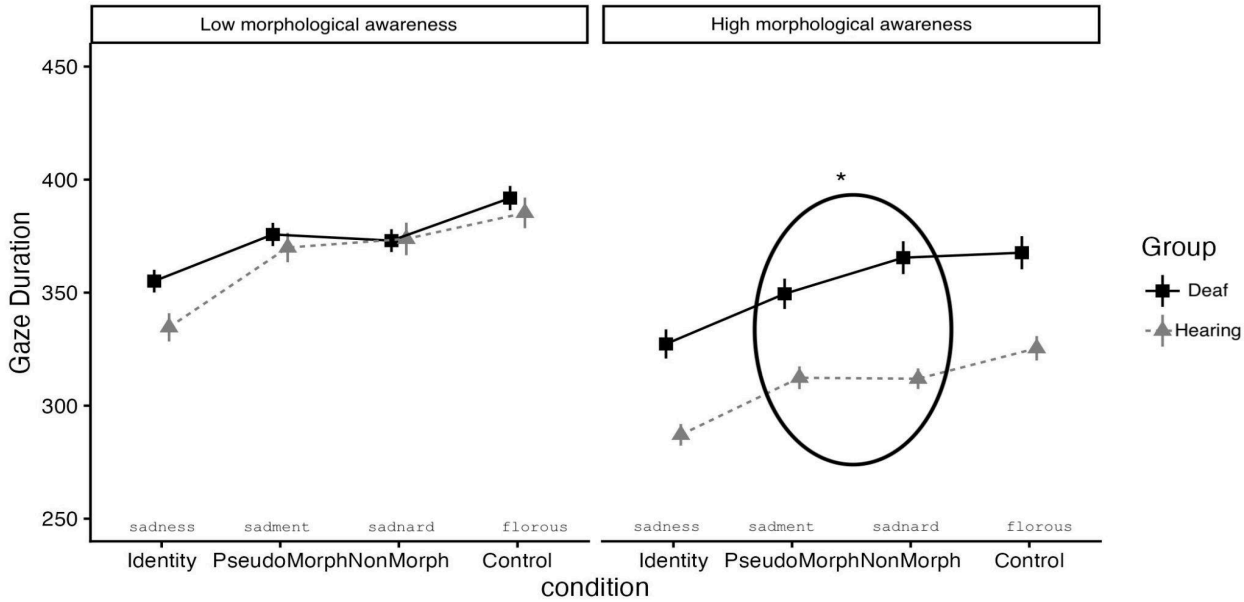

Figure 2. Interaction between group, morphological awareness, and preview condition.

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